Fiscal Year:	FY 2026	**Task Last Updated:**	FY 03/24/2026
PI Name:	Nusbaum, Derek M.D., Ph.D.		
Project Title:	Flight Study – GLP-1 Agonist as Treatment for Clinically Significant SANS		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	KBR/NASA Johnson Space Center		
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City:	Texas	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: Principal Investigator (PI) is also affiliated with the University of Texas Medical Branch.		
Project Type:	Flight	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2025	**End Date:**	09/30/2030
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Boothe, Grace (KBR/NASA Johnson Space Center) Laurie, Steven Ph.D. (KBR/NASA Johnson Space Center) Macias, Brandon Ph.D. (NASA Johnson Space Center) Brunstetter, Tyson O.D., Ph.D. (NASA Johnson Space Center) Mason, Sara (Aegis/NASA Johnson Space Center) Tarver, William M.D., Ph.D. (NASA Johnson Space Center) Smith, Jennyffer O.D., Ph.D. (KBR/NASA Johnson Space Center) Young, Millennia Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Spaceflight-associated Neuro-ocular Syndrome (SANS) is a condition that is hypothesized to occur due to the lack of hydrostatic pressure in weightlessness. This condition can lead to ocular and brain structural changes. Currently, there are no countermeasures proven to treat or prevent the development or worsening of SANS during long-duration spaceflight. The GLP-1 receptor agonist (RA) Exenatide demonstrated the ability to lower intracranial pressure (ICP), likely by decreasing choroid plexus cerebrospinal fluid (CSF) production, in patients with idiopathic intracranial hypertension (IIH) in a phase II clinical trial . It is unknown what role elevated ICP plays in the development of SANS findings, and it is also unknown whether pharmaceutically lowering ICP would treat SANS findings. However, chronically elevated ICP of moderate levels during weightlessness may be responsible for the moderate optic disc edema observed in 70% of long-duration flyers. Therefore, Exenatide represents a therapeutic option that may lower ICP sufficiently to counter spaceflight-induced optic disc edema and other SANS effects. This study will enroll crewmembers flying long-duration spaceflight missions on the International Space Station. The current proposal outlines the use of Exenatide as a rescue treatment for clinically concerning SANS development, as well as the efficacy and safety of GLP1-RA in long-duration spaceflight. All crewmembers will be allowed to consent to participate in this study. Once consented, any crewmember who meets the primary inclusion criteria for clinically concerning SANS will be offered the GLP-1 receptor agonist Exenatide and outcome measures will be tracked for the remainder of their mission.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New Project for FY2026
Bibliography Type:	Description: (Last Updated:)